

E-COMMERCE SUPPLY CHAIN OPTIMIZATION

End-to-End Production Dashboard Deployment Notes

1. Project Objective & Strategic Vision

This project focused on transforming unorganized, flat e-commerce logistics spreadsheets into an enterprise-scale relational analytical tracking application. The dashboard was engineered from the ground up to empower executive operations managers to monitor top-line regional sales volumes, track financial tax liabilities, detect supply-chain fulfillment delays, and isolate high-risk orders with zero friction.

2. Technical Architecture & Implementation Steps

01 Phase 1: Power Query ETL Data Pipeline Engineering

- **Data Transformation:** Cleaned raw inputs inside the Power Query staging workspace. Standardized localized text anomalies across categorical column vectors and re-cast system data types into exact Whole Numbers, Decimal Currencies, and Standard Dates.
- **Conditional Logic Injection:** Developed an automated supply chain tracking system by engineering a custom conditional multi-nested classification layer (Shipping_Status) based on logistics fulfillment performance:

```
if [Delivery days] > 200 then "Critical Delay"
else if [Delivery days] >= 190 then "High Alert"
else "Standard Shipping"
```

02 Phase 2: Relational Schema Modeling

- **Data Decoupling Sequence:** Broke apart the initial massive flat data table to form a production-grade **Star Schema Data Model**, eliminating structural data redundancies.
- **Table Segregation:** Segmented the model into a central transactional metric core named Fact_sales, surrounded by three distinct independent dimension lookup layers: Dim_Customer, Dim_ProductTable, and Dim_region.
- **Relationship Wiring:** Established high-performance single-directional **1-to-Many Relationships** by connecting structural primary index indicators from the lookup tables directly to the active foreign keys inside the central transactional table.

03 Phase 3: Explicit DAX Measurement Programming

Programmed an automated custom calendar master table to establish a global corporate timeline baseline across all database tables: `Calendar = CALENDARAUTO()`. Linked the structural timeline fields to the core transactional table keys to activate advanced explicit mathematical calculations:

- **Fulfillment Speed Tracker:** `Avg Delivery Days = AVERAGE(Fact_sales[Delivery days])`
- **Dynamic Tax Calculator:** `Estimated GST = SUM(Fact_sales[Net sales]) * 0.18`
- **Cumulative Growth Tracker:** `YTD Sales = TOTALYTD(SUM(Fact_sales[Net sales]), 'Calendar'[Date])`
- **Prior Year Analysis Engine:** `Sales Last Year = CALCULATE(SUM(Fact_sales[Net sales]), SAMEPERIODLASTYEAR('Calendar'[Date]))`

04 Phase 4: Executive UI/UX Workspace Design

- **Theme Architecture:** Replaced standard canvas styles with a polished, high-contrast dark corporate executive workspace layout theme to increase long-term data scannability.
- **Fulfillment Alerts Configuration:** Integrated **Exceptions Reporting** directly onto the top-level Avg_Delivery_Days operational tracking card using precise, unit-corrected numerical formatting boundaries (Green ● for performance < 195 days; flashing Alert Red ● for shipping bottlenecks ≥ 195 days).
- **Interactive Analytics Ecosystem:** Synchronized interactive summary blocks (Donut Charts tracking payment methods and fulfillment risk shares) with multi-axis bar charts and multi-layered breakdown matrices to enable seamless data cross-filtering across all visuals.
- **Control Interface:** Converted basic checkbox structures into responsive application ****Tile Slicer Buttons**** (Bulk_Order vs. Retail_sales) to provide clean filter controls for executive stakeholders.

3. The Professional Portfolio Synopsis

This completed project configuration stands as an institutional-grade proof-of-concept demonstrating modern data warehousing, explicit functional coding, and highly responsive user-interface engineering. It proves an analyst's ability to take raw business files and scale them cleanly into complex analytical data products.

Professional Resume Project Bullet Description:

"Architected an interactive dark-themed E-Commerce Supply Chain Optimization Application in Power BI Desktop utilizing an optimized Star Schema data architecture. Engineered clean background ETL processing pipelines within Power Query to transform raw records, enforce type schemas, and ingest multi-nested conditional tracking columns. Segmented loose flat tables into an integrated data model composed of a central fact ledger and three lookup dimension tables bound via single-directional 1-to-Many cardinality pathways. Authored custom mathematical business logic and multi-layered calendar tracking parameters via explicit DAX measures (incorporating advanced context filters and Time Intelligence functions). Constructed a synchronized UI interface using tile slicers, detailed matrix displays, and rule-driven conditional formatting color thresholds to automate operational exceptions reporting."